

CSE 4417 IoT Theory

Lecture 06.5: Controllers (Raspberry Pi)

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Raspberry Pi

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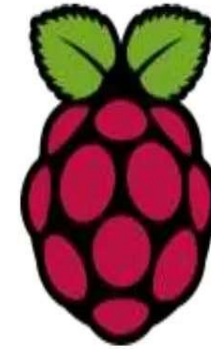
Wireless temperature monitoring system using Pi

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Introduction to Raspberry Pi

Raspberry Pi is defined as a minicomputer the size of a credit card that is interoperable with any input and output hardware device like a monitor, a television, a mouse, or a keyboard – effectively converting the set-up into a full-fledged PC at a low cost.

- **Affordable:** Prices start as low as \$5 for the Raspberry Pi Pico and increase depending on model and capabilities.
- **Compact:** Pocket-sized design (typically around the size of a credit card) makes it portable and convenient.
- **Powerful:** Can handle diverse tasks with processors ranging from single-core to quad-core and RAM from 256MB to 8GB.
- **Versatile:** Offers various input/output ports, including HDMI, USB, and GPIO pins for connecting to different devices and sensors.
- **Open-source:** Supported by a large and active community and runs on open-source operating systems like Raspberry Pi OS (based on Linux).

<https://www.raspberrypi.com/documentation/computers/getting-started.html#setting-up-your-raspberry-pi>

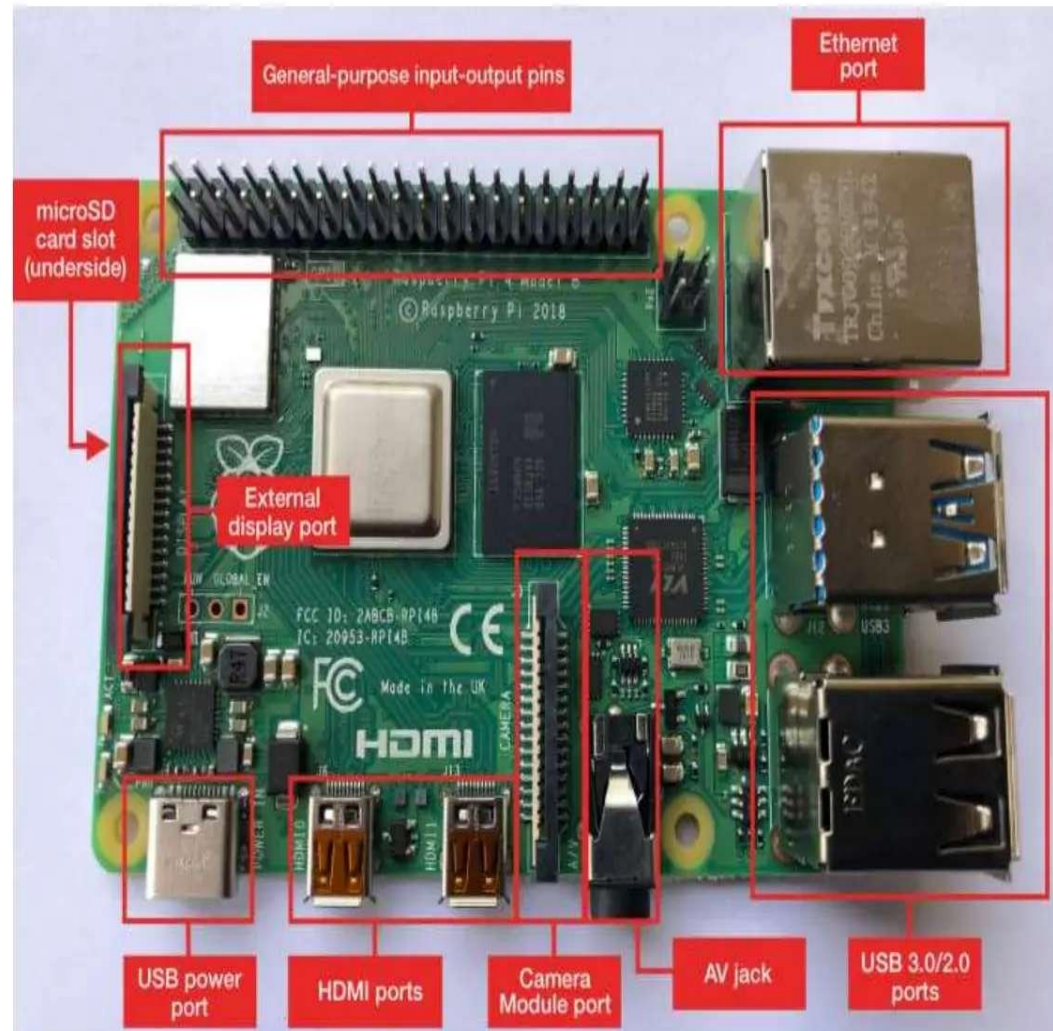
Raspberry Pi and Arduino Difference

Feature	Raspberry Pi	Arduino
Type of device	Microcomputer	Microcontroller
Processing power	More powerful	Less powerful
Operating system	Runs a full operating system, such as Raspberry Pi OS or Ubuntu	Doesn't run an operating system
Programming language	Python, Java, C++, Scratch	C++
Connectivity	Wi-Fi, Bluetooth, Ethernet	Limited to USB
Cost	More expensive	Less expensive
Complexity	More complex to set up and use	Simpler to set up and use
Best for	Complex projects that require multitasking and internet connectivity	Simple projects that require controlling hardware

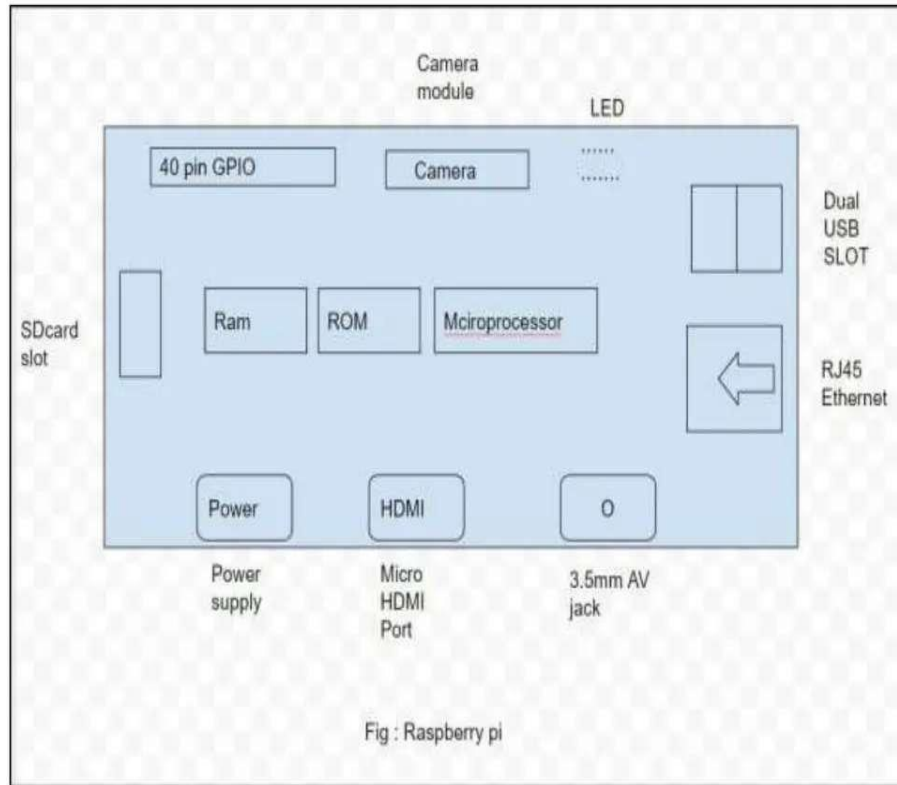
Raspberry Pi and Arduino Difference

- Processing power:** Raspberry Pi has a more powerful processor, which means it can handle more complex tasks, such as running multiple programs at the same time or processing large amounts of data. Arduino has a simpler processor, which is better suited for smaller, more focused tasks.
- Operating system:** Raspberry Pi runs a full operating system, which gives you more flexibility and control over your projects. Arduino doesn't run an operating system, so you have less control over how it works.
- Programming language:** Raspberry Pi can be programmed in a variety of languages, including Python, Java, and C++. Arduino can only be programmed in C++.
- Connectivity:** Raspberry Pi has built-in Wi-Fi, Bluetooth, and Ethernet connectivity, which makes it easy to connect to the internet and other devices. Arduino typically only has USB connectivity.
- Cost:** Raspberry Pi is more expensive than Arduino. However, it is also more powerful and versatile.
- Complexity:** Raspberry Pi is more complex to set up and use than Arduino. This is because it runs an operating system and has more features. However, there are many resources available to help you get started with Raspberry Pi.
- Best for:** Raspberry Pi is best for complex projects that require multitasking and internet

Raspberry Pi board : Hardware layout



Raspberry Pi board : Hardware layout



<https://www.youtube.com/watch?v=oWxoeU9v3K8>

<http://meseec.ce.rit.edu/551-projects/spring2017/2-3.pdf>

Raspberry Pi board : Hardware layout

System on a Chip (SoC):

This is the brain of the Raspberry Pi, integrating the Central Processing Unit (CPU), Graphics Processing Unit (GPU), and Random Access Memory (RAM) onto a single chip. Different models boast varying processing power and graphics capabilities.

Storage:

The Raspberry Pi relies on a microSD card for storing its operating system and files. Choose a high-quality card with sufficient capacity for your needs, as it plays a crucial role in performance and data safety.

Power Supply:

A stable power supply is vital for smooth operation. Most models require a Micro USB or USB-C power supply that can deliver at least 2.5 amps of current. Choose a reliable power adapter to avoid any hiccups.

Input/Output (I/O) Ports:

- HDMI: for connecting to a monitor or TV
- USB: for connecting peripherals like keyboards, mice, and external storage
- Ethernet: for wired network connection
- 3.5mm jack: for audio output
- MicroSD card slot: for storage expansion



Raspberry Pi board : Hardware layout

General-Purpose Input/Output (GPIO) Pins:

These 40 pins on the Raspberry Pi's header offer immense flexibility. You can connect various sensors, actuators, and other devices to interact with the physical world, enabling exciting projects like robotics, home automation, and more.

3V3	3.3 volts	Anything connected to these pins will always get 3.3V of power
5V	5 volts	Anything connected to these pins will always get 5V of power
GND	ground	Zero volts, used to complete a circuit
GP2	GPIO pin 2	These pins are for general-purpose use and can be configured as input or output pins
ID_SC/ID_SD/DNC	Special purpose pins	



Operating Systems on Raspberry Pi

Raspberry Pi needs optimized OS suitable to run on Raspberry Pi hardware.

Raspberry Pi OS (formerly Raspbian)

The official and most popular choice, based on Debian Linux. Beginner-friendly with a desktop environment and pre-installed software. Available in 32-bit and 64-bit versions.

Ubuntu Core

Another Linux-based option, lightweight and focused on security and minimal resource usage. Ideal for Internet of Things (IoT) projects and headless servers.

LibreELEC and OSMC

These media center distributions turn your Pi into a dedicated streaming device, supporting popular services like Netflix, Kodi, and Plex.

Raspberry Pi OS

Beginner-friendly: Raspberry Pi OS boasts a familiar desktop interface with pre-installed software for everyday tasks like browsing the web, watching videos, editing documents, and coding. You don't need prior Linux experience to get started

- **Lightweight and efficient:** Optimized for Raspberry Pi's hardware, the OS runs smoothly even on models with limited resources, making it ideal for resource-intensive projects like robotics or IoT applications.
- **Highly customizable:** With access to the vast Debian package repository, you can install countless software applications to tailor the OS to your specific needs and interests.
- **Educational focus:** Raspberry Pi OS includes educational software like Scratch and Python, making it a fantastic platform for learning programming and electronics.
- **Community-driven:** A vibrant community of Raspberry Pi enthusiasts and developers provides extensive support and resources, ensuring you're never alone in your Raspberry Pi journey.

https://en.wikipedia.org/wiki/Raspberry_Pi_OS

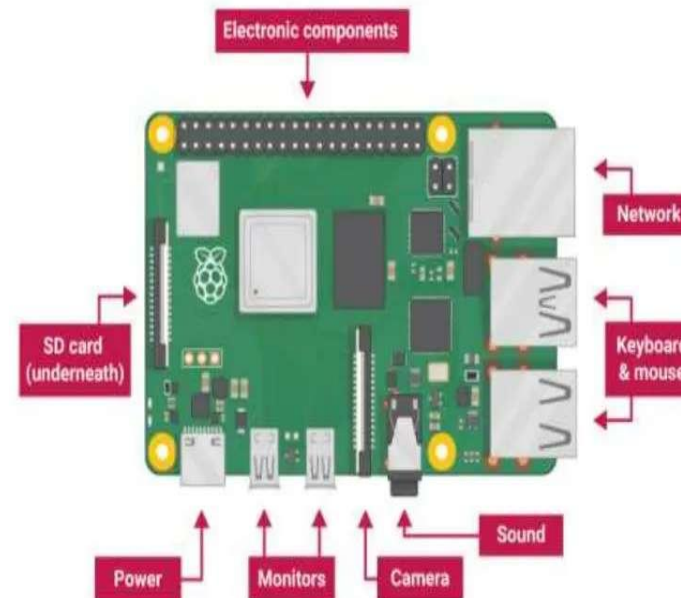
Configuring Raspberry Pi

Pre-requisites

- Power Supply
- microSD Card
- Keyboard / Mouse
- TV/Monitor Screen

Steps

- Setup SD Card and install OS
- Connect Raspberry Pi
- Powerup
- Finish Installation



<https://projects.raspberrypi.org/en/projects/raspberry-pi-setting-up>

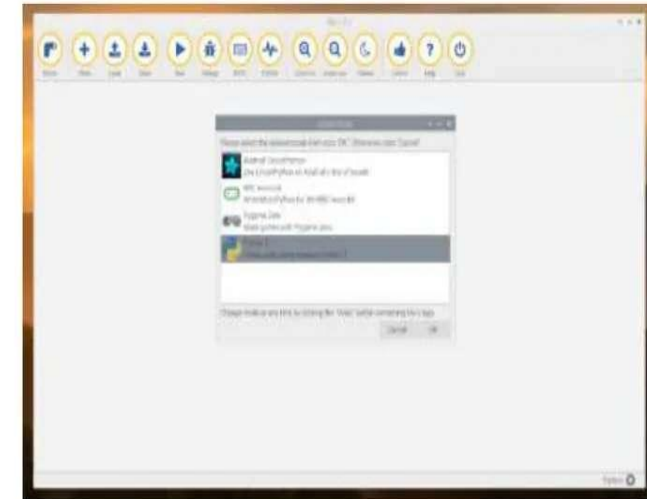
Programming Raspberry Pi with Python

- Python comes preinstalled on Raspbian
- The Raspbian operating system comes with several preinstalled Python IDEs that you can use to write can use to write your programs. One of these IDEs is [Mu](#). It can be found in the main menu:
menu:

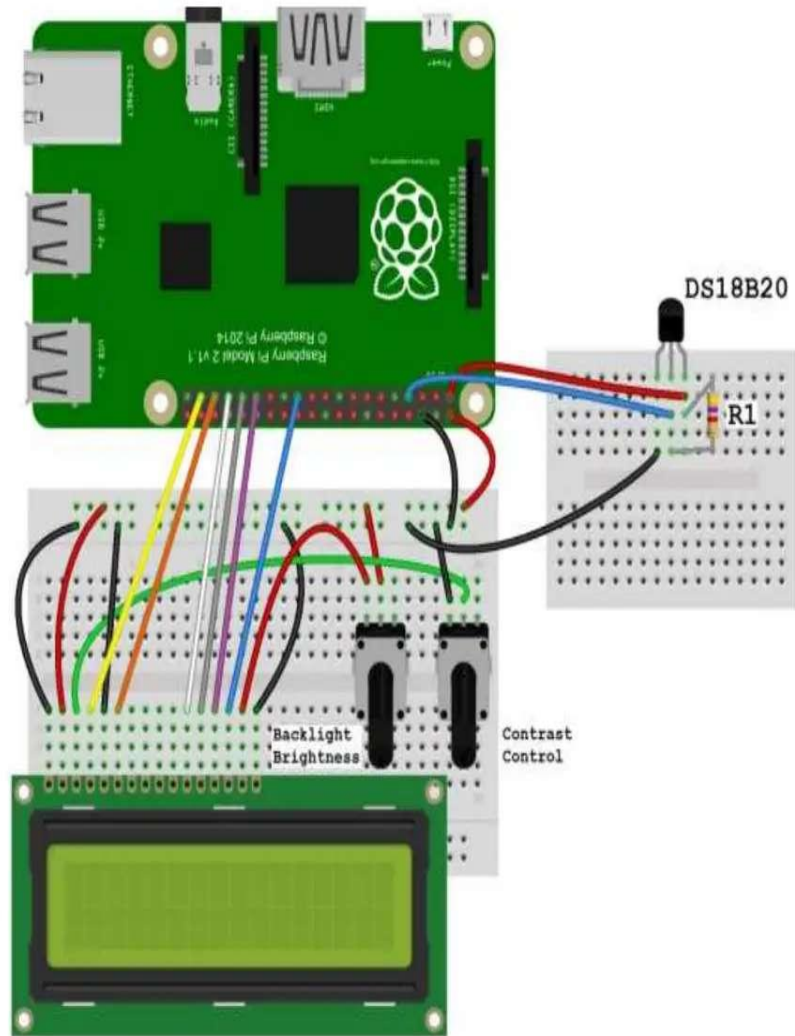
Raspberry Pi Icon → Programming → Mu

Steps

1. Open Mu by going to *Raspberry Pi Icon → Programming → Mu*.
2. Click *New* in the menu bar to create an empty file.
3. Click *Save* in the menu bar.
4. Navigate to the `/home/pi` directory in the directory dropdown.
5. Click the *Create New Folder* icon in the top-right corner.
6. Name this new directory `python-projects` and hit Enter.
7. Click *Cancel* to close.



Wireless temperature monitoring system using Pi



DS18B20 Temperature Sensor

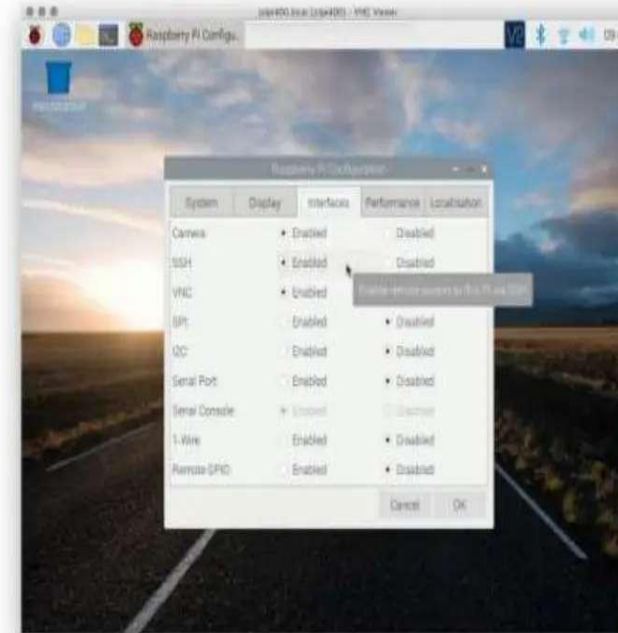
No:	Pin Name	Description
1	Ground	Connect to the ground of the circuit
2	Vcc	Powers the Sensor, can be 3.3V or 5V
3	Data	This pin gives output the temperature value which can be read using 1-wire method



Connecting Raspberry Pi via SSH

- Enable SSH on Raspberry Pi
- Connecting via SSH
 - Find out IP Address
 - `hostname -I`
 - `ssh username@IPAddress`

<https://raspberrypi-guide.github.io/networking/connecting-via-ssh>



Accessing temperature from DS18B20 sensors

```
import os
import glob
import time

os.system('modprobe w1-gpio')
os.system('modprobe w1-therm')

base_dir = '/sys/bus/w1/devices/'
device_folder = glob.glob(base_dir + '28*')[0]
device_file = device_folder + '/w1_slave'

def read_temp_raw():
    f = open(device_file, 'r')
    lines = f.readlines()
    f.close()
    return lines

def read_temp():
    lines = read_temp_raw()
    while lines[0].strip()[-3:] != 'YES':
        time.sleep(0.2)
        lines = read_temp_raw()
    equals_pos = lines[1].find('t=')
    if equals_pos != -1:
        temp_string = lines[1][equals_pos+2:]
        temp_c = float(temp_string) / 1000.0
        temp_f = temp_c * 9.0 / 5.0 + 32.0
        return temp_c, temp_f

while True:
    print(read_temp())
    time.sleep(1)
```

Load the following kernel modules (w1-gpio and w1-therm) to interface with the DS18B20 sensor via one-wire communication protocol.

The function `read_temp_raw()` gets the temperature readings from the `w1_slave` file (that's the place where those are stored). This function opens the `w1_slave` file, reads its contents line by line, and then returns a list containing the lines.

Reads the actual data

Remote Access to Raspberry Pi

Enable Remote Access:

- SSH (Secure Shell): This is the most common method for command-line access. It's typically enabled by default in Raspberry Pi OS.

- To verify, open a terminal and type `sudo raspi-config`.
- Navigate to "Interfacing Options" > "SSH" and enable it if necessary.

- VNC (Virtual Network Computing): This allows for full graphical desktop access remotely.

- Go to "Menu" > "Preferences" > "Raspberry Pi Configuration" > "Interfaces" and enable VNC.
- Set a password for VNC access.

2. Find Your Raspberry Pi's IP Address:

- Open a terminal and type `hostname -I` to display its IP address on your local network.
- Note this address for connecting from other devices.

3. Choose Your Remote Access Method:

A. SSH (Command-Line Access):

- From a Linux or macOS computer: Open a terminal and type `ssh pi@<IP_address>` (replace <IP_address> with your Pi's actual IP).

- From a Windows computer: Use an SSH client like PuTTY.

B. VNC (Graphical Desktop Access):

- Install a VNC viewer app on your device (e.g., VNC Viewer, RealVNC Viewer).
- Enter your Pi's IP address and VNC password when prompted.

C. Third-Party Remote Access Solutions:

- Chrome Remote Desktop: Simple setup for remote desktop access from anywhere (requires a Google account).
- TeamViewer: Versatile option for both desktop and command-line access, often used for professional support.

References

- IoT for data science & Analytics - Sourav Tripathy



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Thank You

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